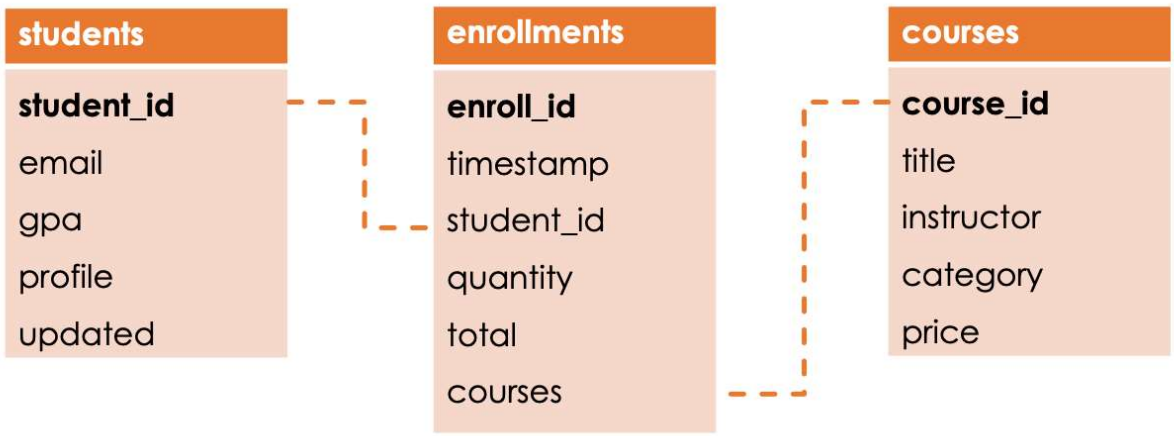
(https://databricks.com)

Lab Solution: implementing a DLT pipeline

This notebook is **not intended** to be executed interactively, but rather to be deployed as a DLT pipeline from the **workflows** tab

- Help: DLT syntax documentation (https://docs.databricks.com/en/delta-live-tables/tutorial-sql.html).



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```
SET datasets.path=dbfs:/mnt/DE-Associate/datasets/school;
```

Q1- Declaring Bronze Tables

Declare a streaming live table, `enrollments_bronze`, that ingests JSON data incrementally using Auto Loader from the directory **"${datasets.path}/enrollments-json-raw"**

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```
-- ANSWER
CREATE OR REFRESH STREAMING LIVE TABLE enrollments_bronze
AS SELECT * FROM cloud_files("${datasets.path}/enrollments-json-raw", "json",
                             map("cloudFiles.inferColumnTypes", "true"))
```

Declare a live table, `students_bronze`, that load data directly from JSON files in the directory **"${datasets.path}/students-json"**

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```
-- ANSWER
CREATE OR REFRESH LIVE TABLE students_bronze
AS SELECT * FROM json.`${datasets.path}/students-json`
```

Q2 - Declaring Silver Table

Declare a streaming live table, `enrollments_cleaned`, that:

1. Enrich the **enrollments_bronze** data through an inner join with the `students_bronze` table on the common `student_id` field to obtain the student's country
2. Implement quality control by applying a constraint to drop records with a null `email`
3. The table will have the following schema:

Field	Type
`enroll_id`	`STRING`
`total`	`DOUBLE`
`email`	`STRING`
`country`	`STRING`

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```
-- ANSWER
CREATE OR REFRESH STREAMING LIVE TABLE enrollments_cleaned (
  CONSTRAINT valid_email EXPECT (email IS NOT NULL) ON VIOLATION DROP ROW
)
AS SELECT enroll_id, total, email, profile:address:country as country
  FROM STREAM(LIVE.enrollments_bronze) n
  LEFT JOIN LIVE.students_bronze s
    ON n.student_id = s.student_id
```

Q3- Declaring Gold Table

Declare a live table, `course_sales_per_country` against `enrollments_cleaned` that calculate per `country` the following:

- `enrollments_count`: the number of enrollments
- `enrollments_amount`: the sum of the total amount of enrollments

Add a comment to the table: "Course Sales Per Country"

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```
-- ANSWER
CREATE OR REFRESH LIVE TABLE course_sales_per_country
COMMENT "Course Sales Per Country"
AS
  SELECT country, count(enroll_id) AS enrollments_count, sum(total) AS enrollments_amount
  FROM LIVE.enrollments_cleaned
  GROUP BY country
```

Q4- Deploying DLT pipeline

From the **Workflows** button on the sidebar, under the **Delta Live Tables** tab, click **Create Pipeline**

Configure the pipeline settings specified below:

Setting	Instructions
Pipeline name	School DLT
Product edition	Choose **Advanced**
Pipeline mode	Choose **Triggered**
Source code	Use the navigator to select this current notebook (4.1L - Delta Live Tables)
Storage location	dbfs:/mnt/DE-Associate/dlt/school
Target schema	DE_Associate_School_DLT
Cluster policy	Leave it **None**
Cluster mode	Choose **Fixed size**
Workers	Enter **0**
Photon Acceleration	Leave it unchecked
Advanced Configuration	Click **Add Configuration** and enter: - Key: **datasets.path** - Value: **dbfs:/mnt/DE-Associate/datasets/school**
Channel	Choose **Current**

Finally, click **Create**.

Q5 - Run your Pipeline

Select **Development** mode and Click **Start** to begin the update to your pipeline's tables